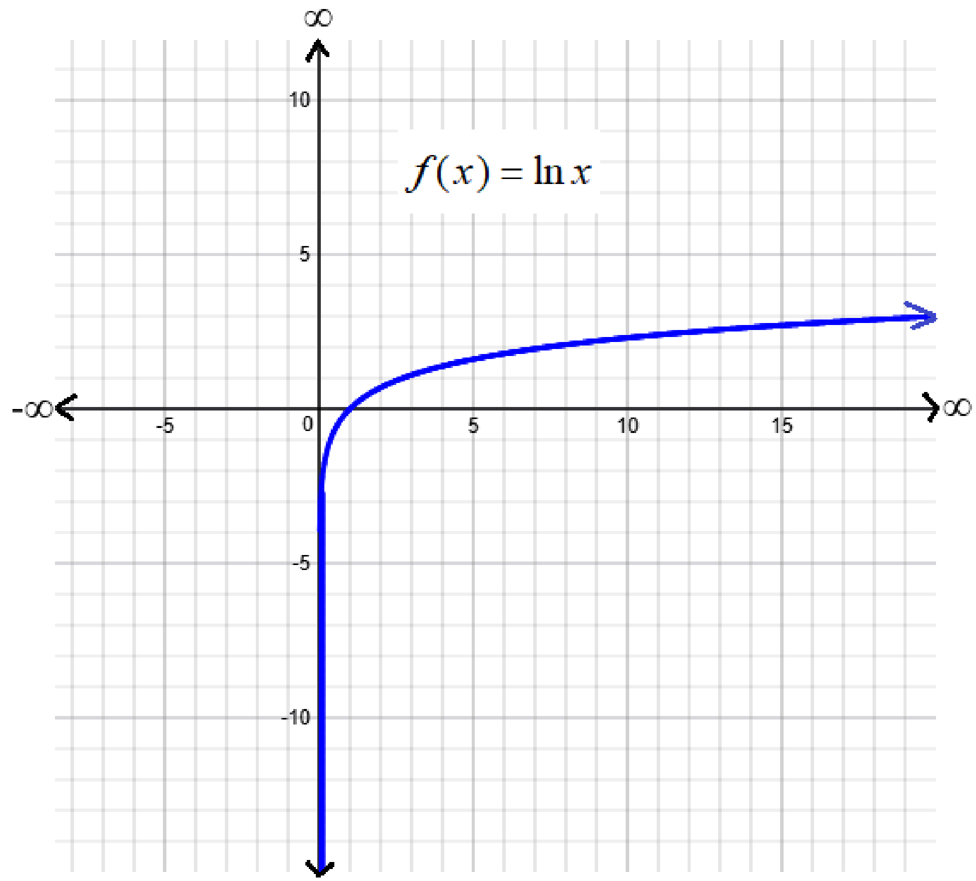
Calculus I Lesson 39B

1) $f(x) = \ln x = \int_1^x \frac{1}{t} \cdot dt, \qquad x > 0$

a) Domain of the function $f(x) = \ln x$ is ___?___

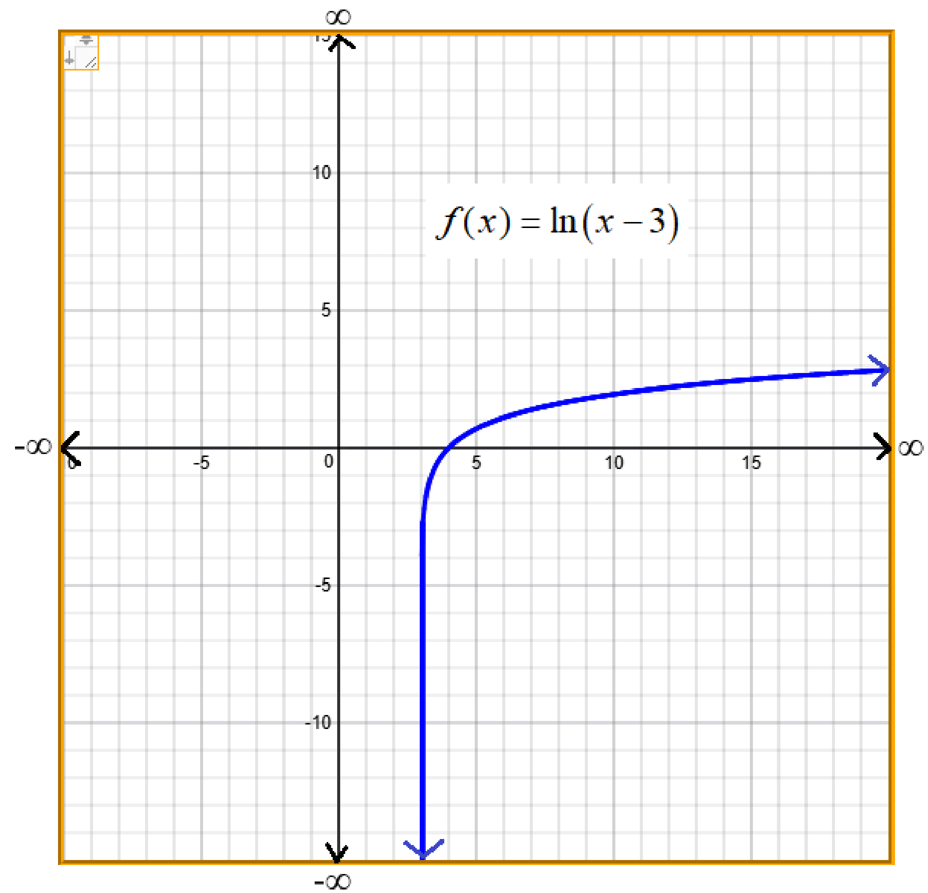
b) Range of the function $f(x) = \ln x$ is ___?___



2) $f(x) = \ln(x-3) = \int_4^x \frac{1}{t-3} \cdot dt, \qquad x > 3$

a) Domain of the function $f(x) = \ln(x-3)$ is ___?___

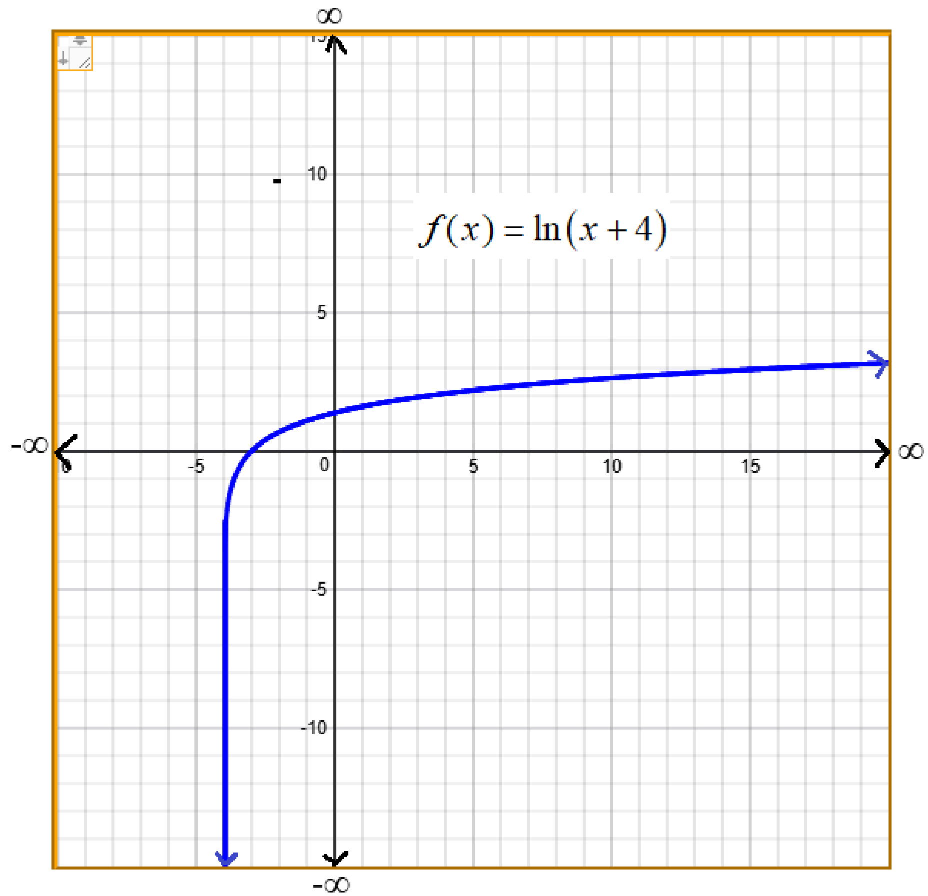
b) Range of the function $f(x) = \ln(x-3)$ is ___?___



3) $f(x)=\ln(x+4)=\int_{-3}^{x}\frac{1}{t+4}\cdot dt,\qquad x>-4$

a) Domain of the function $f(x)=\ln(x+4)$ is ___?___

b) Range of the function $f(x)=\ln(x+4)$ is ___?___



4) a) $\ln(3 \cdot 4) =$ ___?___ b) $\ln(3) + \ln(4) =$ ___?___

5) a) $\ln(7 \cdot 12) =$ ___?___ b) $\ln(7) + \ln(12) =$ ___?___

6) a) $\ln(6 \cdot 14) =$ ___?___ b) $\ln(6) + \ln(14) =$ ___?___

7) $\ln(x \cdot y) =$ ___?___

8) a) $\ln\left(\frac{3}{4}\right) =$ ___?___ b) $\ln(3) - \ln(4) =$ ___?___

9) a) $\ln\left(\frac{7}{12}\right) =$ ___?___ b) $\ln(7) - \ln(12) =$ ___?___

10) a) $\ln\left(\frac{6}{14}\right) =$ ___?___ b) $\ln(6) - \ln(14) =$ ___?___

11) $\ln\left(\frac{x}{y}\right) = \underline{\quad ? \quad}$

12) a) $\ln(3^2) = \underline{\quad ? \quad}$

b) $2 \cdot \ln(3) = \underline{\quad ? \quad}$

13) a) $\ln(7^3) = \underline{\quad ? \quad}$

b) $3 \cdot \ln(7) + \ln(12) = \underline{\quad ? \quad}$

14) a) $\ln(6^4) = \underline{\quad ? \quad}$

b) $4 \cdot \ln(6) = \underline{\quad ? \quad}$

15) $\ln(x^a) = a \cdot \underline{\quad ? \quad}$

Properties of the Natural Log Function:

a) $\ln(x \cdot y) = \ln x + \ln y$

b) $\ln\left(\dfrac{x}{y}\right) = \ln x - \ln y$

c) $\ln\left(x^a\right) = a \cdot \ln x$

16) Expand $\ln\left(x \cdot (x-5)^2\right)$.

$$\ln\left(x \cdot (x-5)^2\right) = \ln(x) + \ln(x-5)^2 = \ln(x) + 2 \cdot (\underline{\quad ? \quad})$$

17) Expand $\ln\left(\sqrt{x \cdot y}\right)$.

$$\ln\left(\sqrt{x \cdot y}\right) = \ln(x \cdot y)^{1/2} = \frac{1}{2} \cdot \ln(x \cdot y) = \frac{1}{2} \cdot [\underline{\quad ? \quad}]$$

18) Expand $\ln\left(x \cdot \sqrt{x^2 + 15}\right)$.

$$\ln\left(x \cdot \sqrt{x^2 + 15}\right) = \ln(x) + \ln\left(\sqrt{x^2 + 15}\right) = \ln(x) + \ln\left(x^2 + 15\right)^{1/2} = \ln(x) + \frac{1}{2} \cdot [\underline{\quad ? \quad}]$$

19) Expand $\ln\left(\sqrt{\dfrac{4x-1}{x}}\right)$.

$$\ln\left(\sqrt{\frac{4x-1}{x}}\right) = \ln\left(\frac{4x-1}{x}\right)^{1/2} = \frac{1}{2}\cdot\left[\ln\left(\frac{4x-1}{x}\right)\right] = \frac{1}{2}\cdot\left[\ln(\underline{\quad ?\quad}) - \ln(\underline{\quad ?\quad})\right]$$

20) Combine $\ln(x)+\ln(x-1)$

$$\ln(x+2)+\ln(x-1) = \ln\left[(x+2)\cdot(x-1)\right] = \ln(\underline{\quad ?\quad})$$